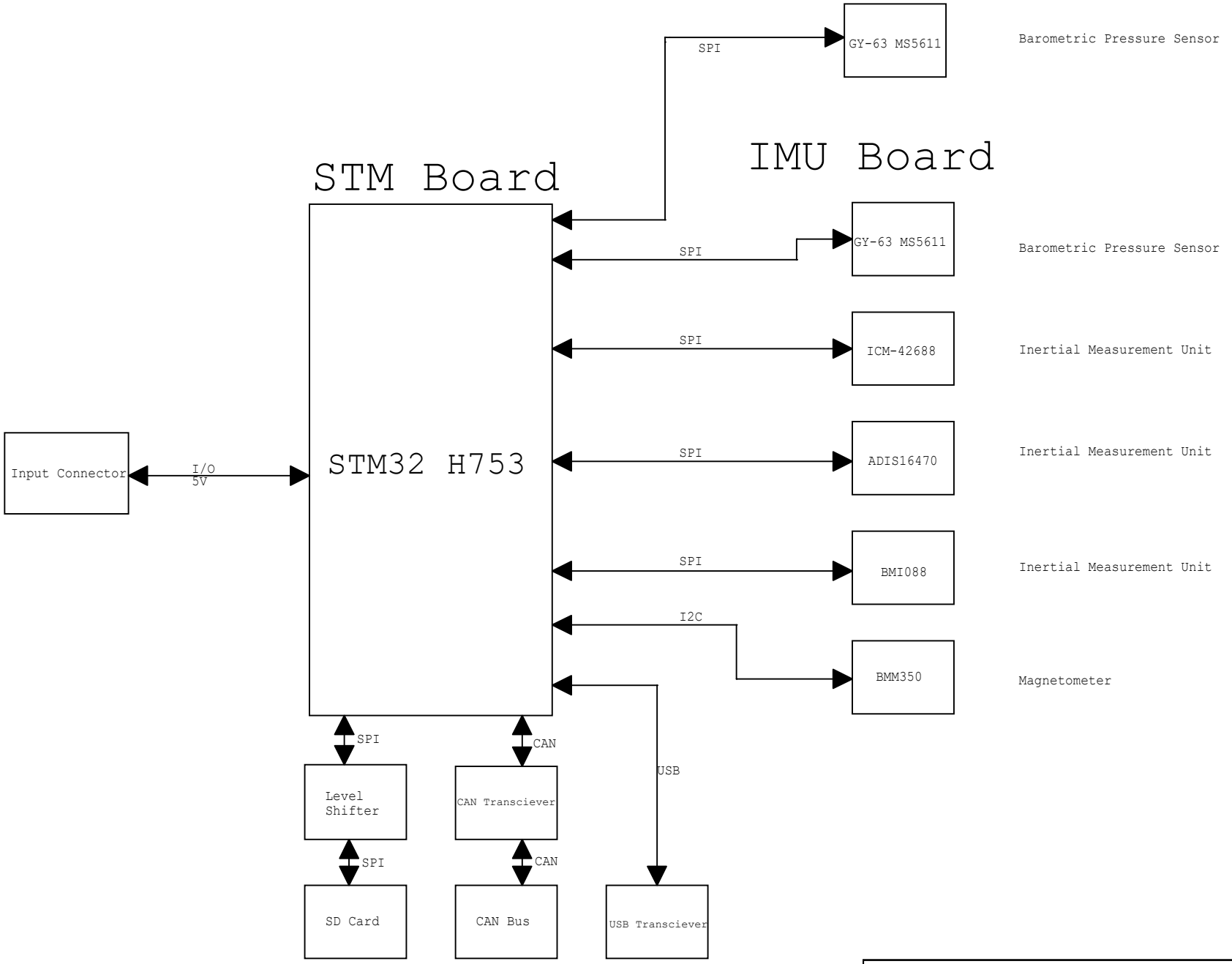
FLIGHT CONTROL BOARD

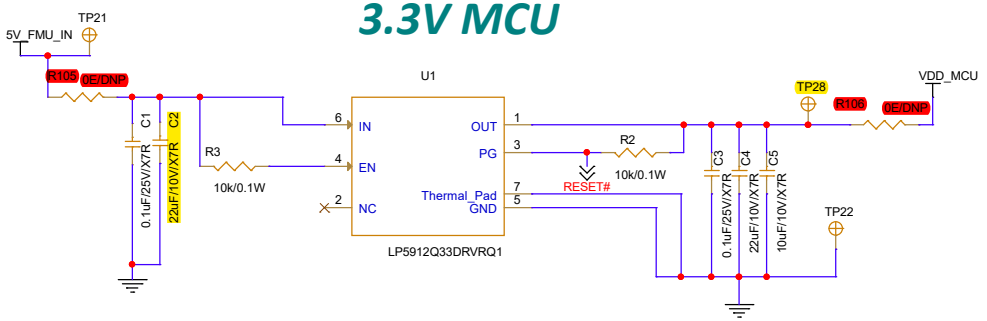
DIGITAL SHARK TECHNOLOGY		
Title: Cover Page		
Size: A	Document Number: 1.0	Rev: 1.1
Date: Friday, November 14, 2025	Sheet 0 of 9	

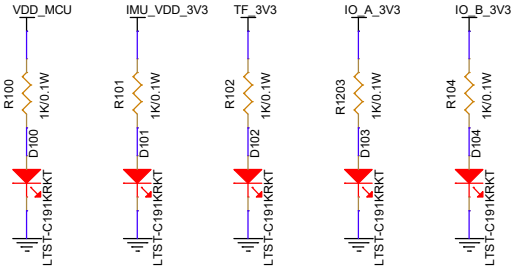


DIGITAL SHARK TECHNOLOGY

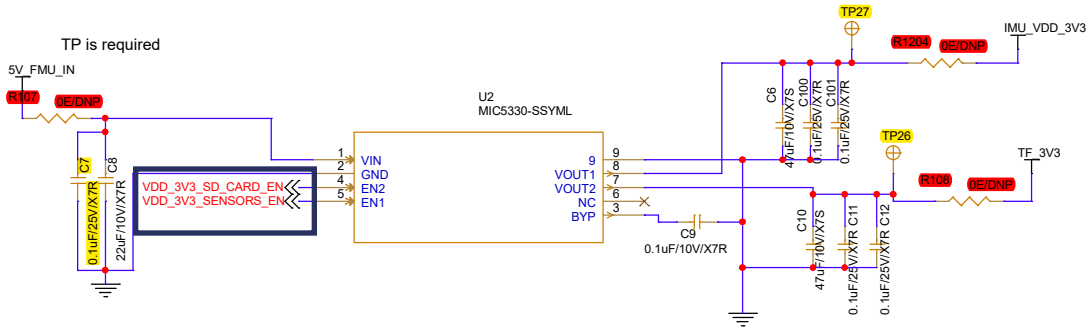
Title: Block Diagram		
Size: A	Document Number: 1.0	Rev: 1.1
Date: Friday, November 14, 2025	Sheet 1 of 9	

3.3V MCU





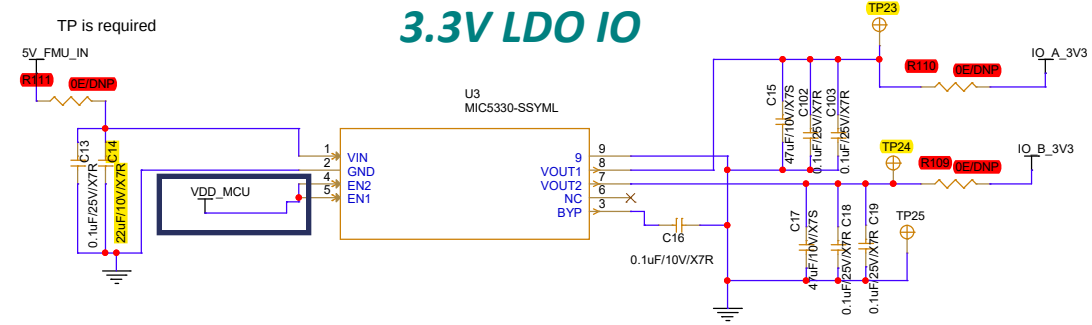
3.3V LDO IMU/TF



Turned- On /Enabled through STM 32 Microcontroller; Need to be measured Later

Voltage Coming from Baseboard

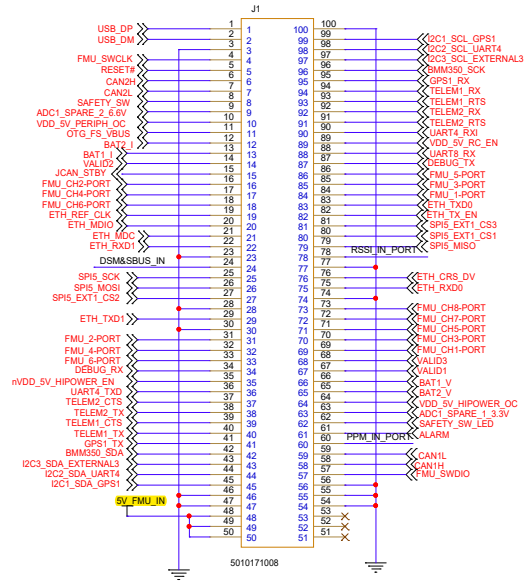
3.3V LDO IO



Schematic Error:
R105, R106, R107, R1204, R108, R111,R110, R109 ==> 0E ==> Should be mounted

DIGITAL SHARK TECHNOLOGY		
Title: Regulators		
Size: B	Document Number: 1.0	Rev: 1.1
Date: Friday, November 14, 2025	Sheet 2 of 9	

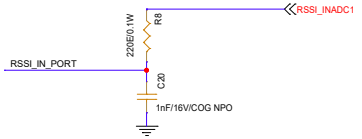
100P CONNECTOR



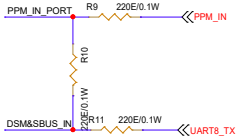
OVER CURRENT



RSSI

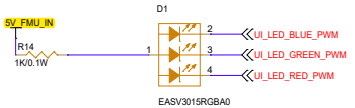


RC IN

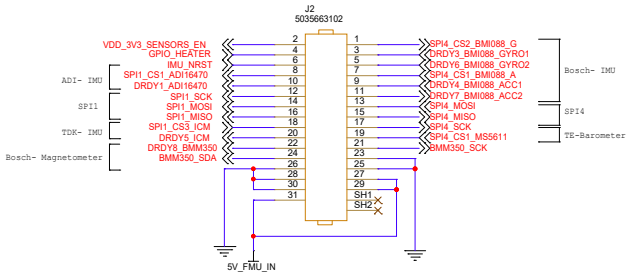


Separate Resistors required for each LED; Resistors values to be calculated

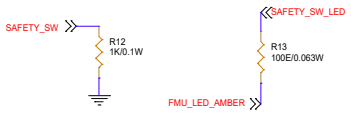
FMU LED

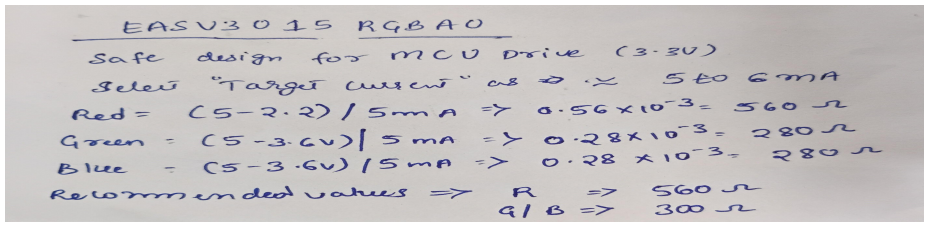


IMU FFC Connector



B to B interface

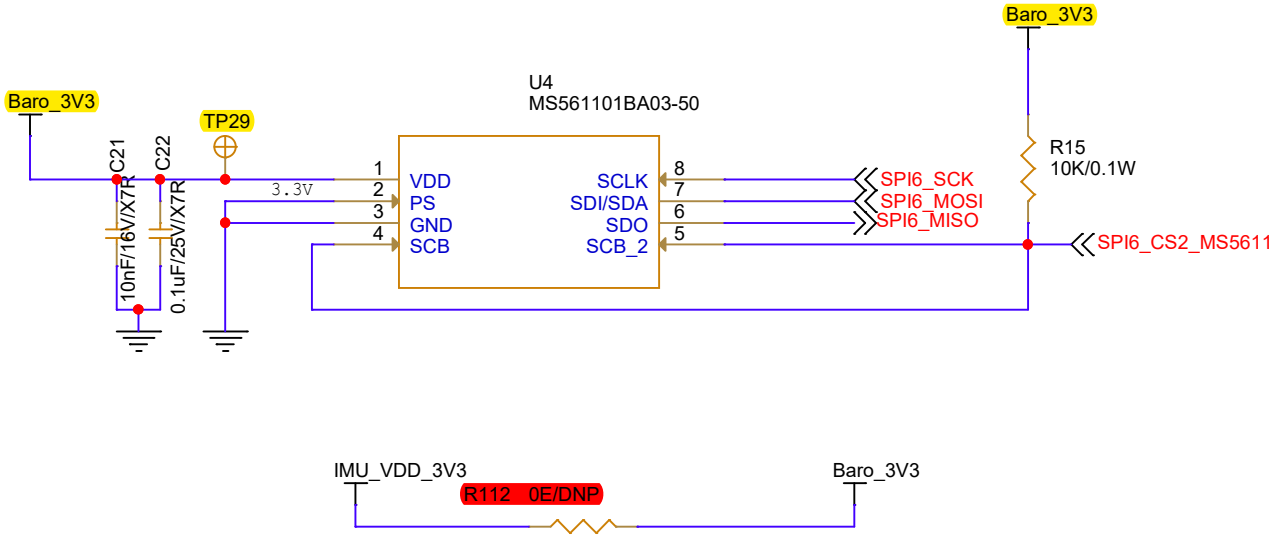




DIGITAL SHARK TECHNOLOGY

Title: Connectors		
Size: C	Document Number: 1.0	Rev: 1.1
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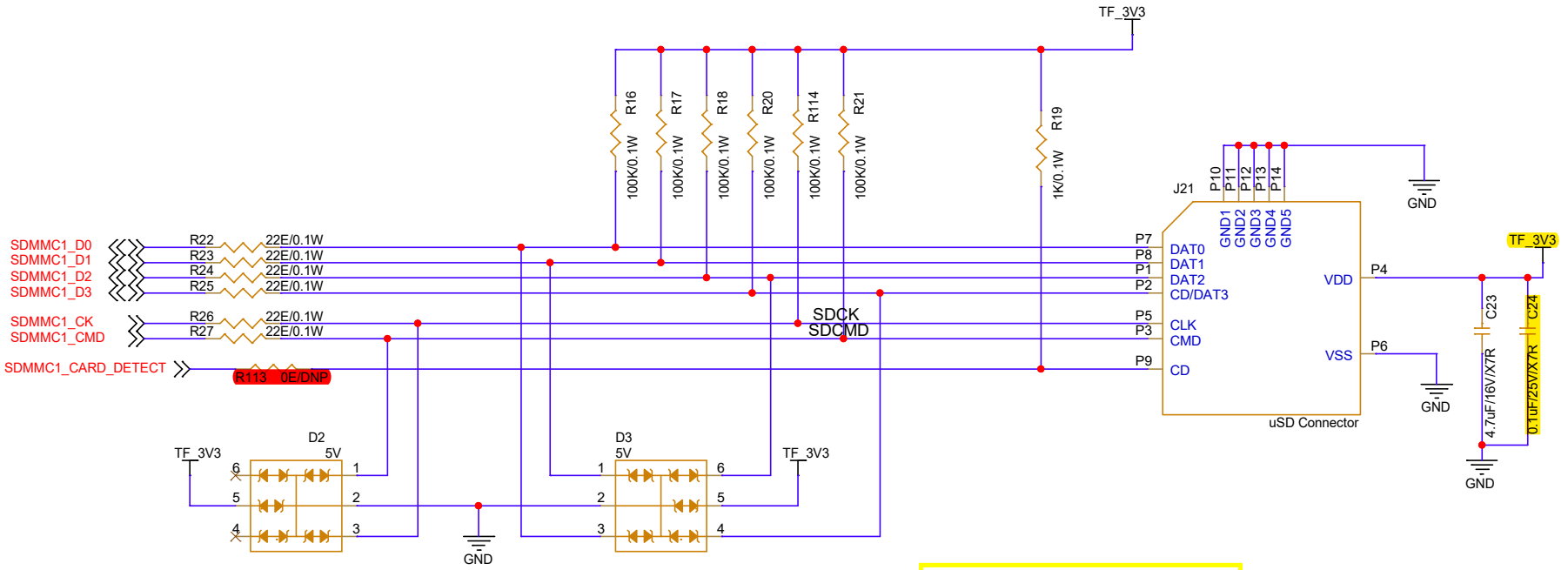
MS5611



DIGITAL SHARK TECHNOLOGY		
Title: IMU		
Size: A	Document Number: 1.0	Rev: 1.1
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SD CARD - FMU

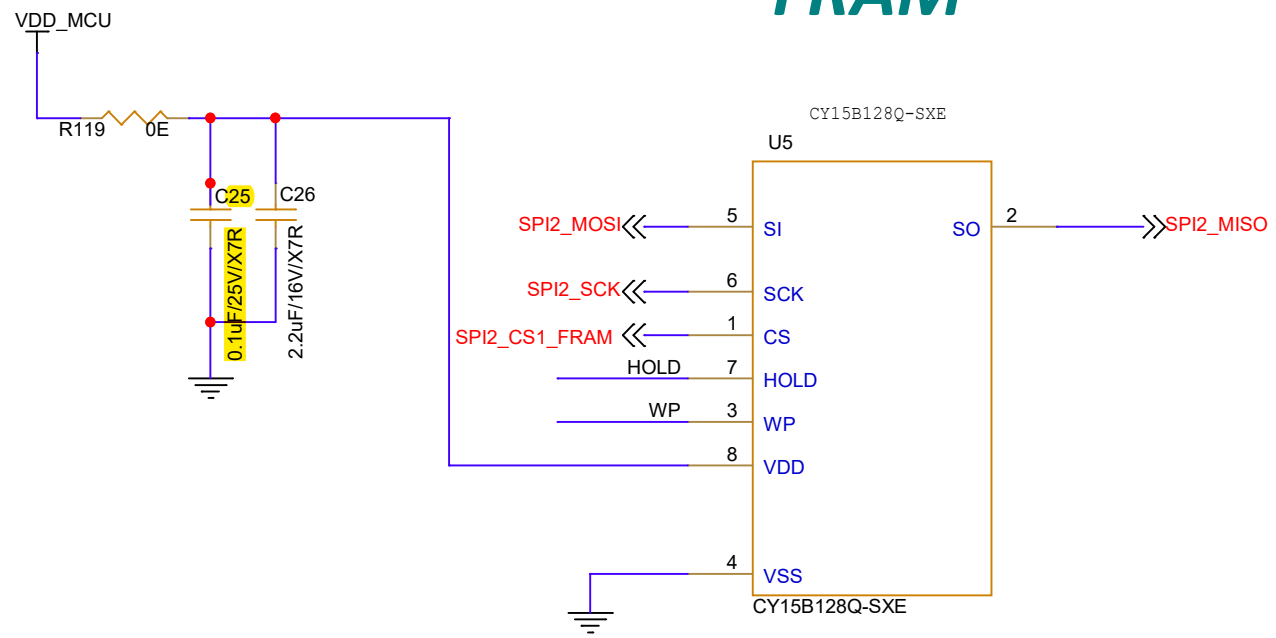
R114 to be removed,
Pull-up is not required; as per EVK
If mounted, SD card detection status will be read incorrectly

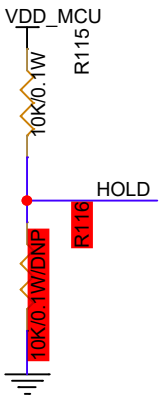


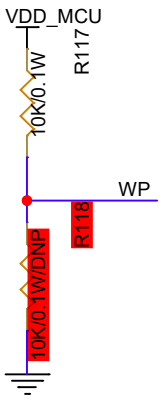
Place the decoupling capacitors
as close as possible to supply pin

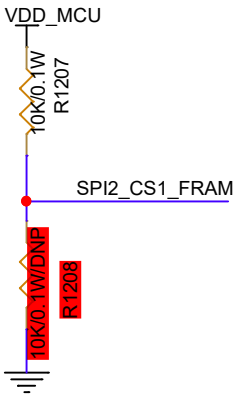
DIGITAL SHARK TECHNOLOGY		
Title: SD Card		
Size: Custom	Document Number: 1.0	Rev: 1.1
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FRAM





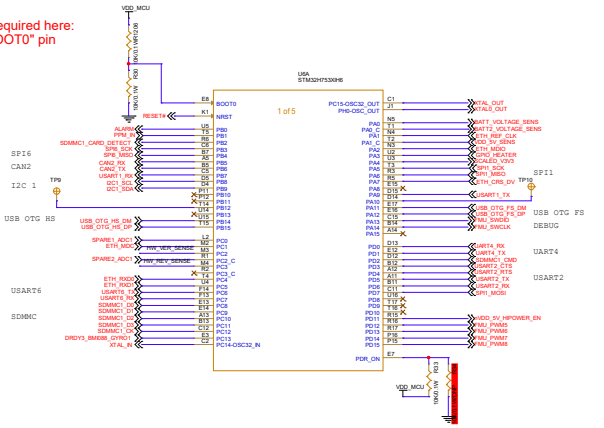


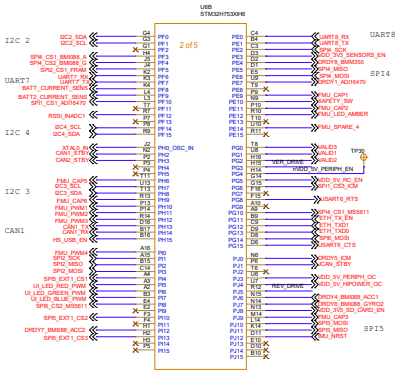


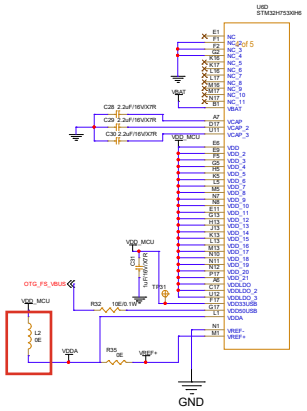
DIGITAL SHARK TECHNOLOGY		
Title: FRAM		
Size: A	Document Number: 1.0	Rev: 1.1
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STM32 H753

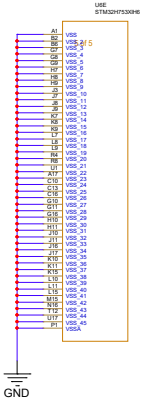
DIP Switch is required here:
For Making "BOOT0" pin
HIGH & LOW.





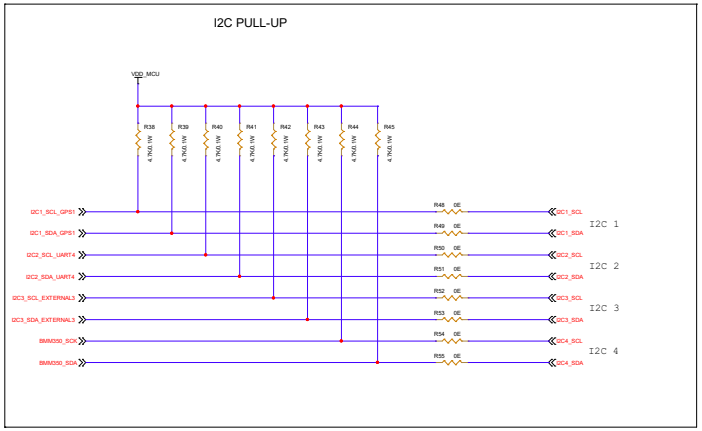




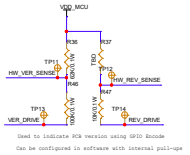


Instead of Resistor - Inductor Symbol
is Placed

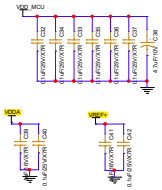
I2C PULL-UP



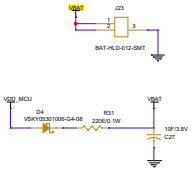
HARDWARE VERSION & ID



DECOUPLING CAPACITORS

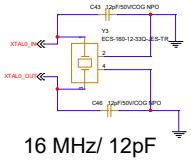


RTC Battery



NOTE:
Higher voltage(3.3V, VDD(MCU)) gets priority
Estimated battery life on only CR2032: 10 Months

CRYSTALS



16 MHz/ 12pF

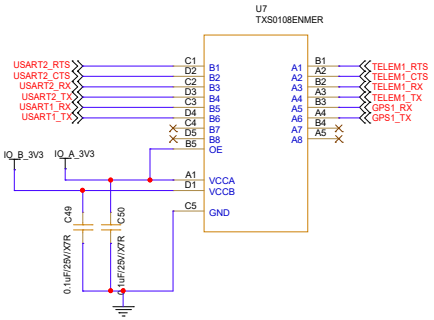


32.768 kHz/ 12.5pF

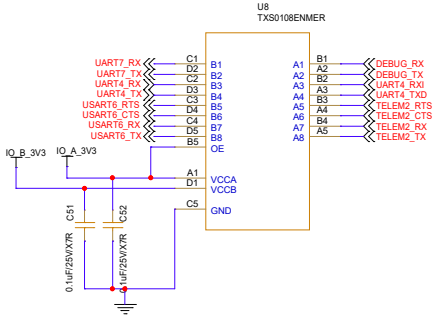
DIGITAL SHARK TECHNOLOGY

Title: STM32 H753			
Size: D	Document Number: 1.0		Rev: 1.1
Date: Friday, November 14, 2008		Sheet 7 of 9	

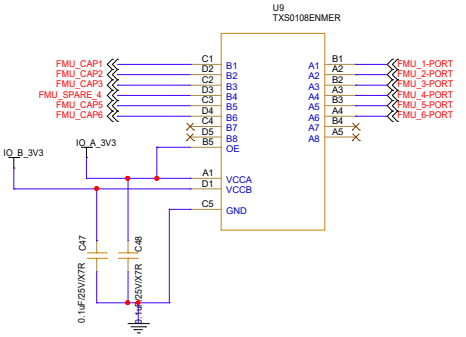
UART



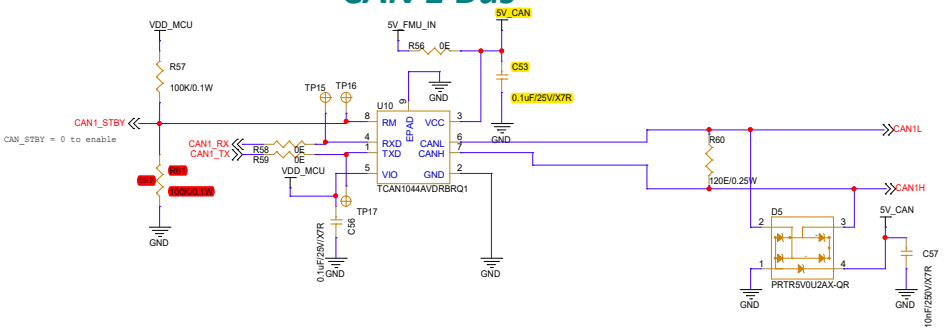
UART



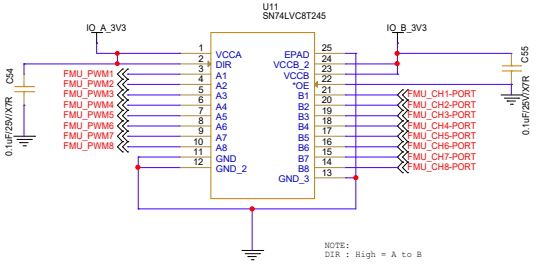
STANDARD PWM



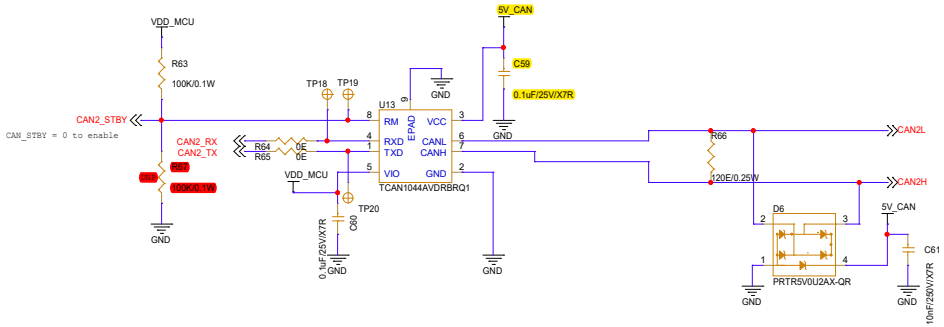
CAN 1 Bus



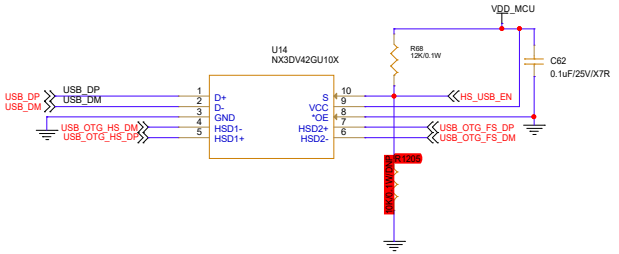
DShot PWM



CAN 2 Bus



USB PORT SWITCH



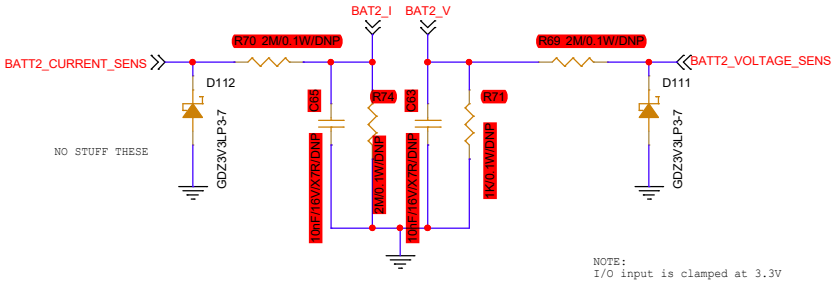
DIGITAL SHARK TECHNOLOGY

Title	Protocol	
Size C	Document Number 1.0	Rev 1.1
Date: Friday, November 14, 2025	Sheet 8 of 9	

PM1 & PM2 Voltage and Current sensing

Battery 2

D105 to D112 - also not required we can make them "DNP" in next version

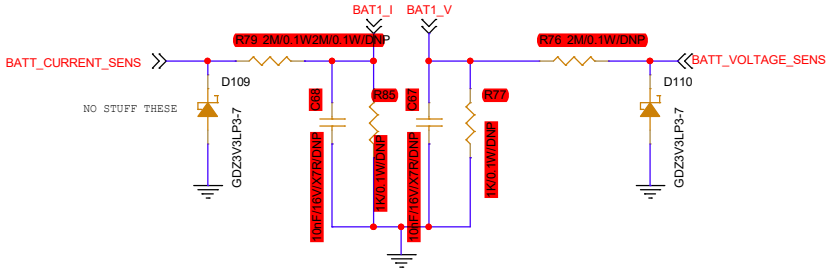


NOTE:
I/O input is clamped at 3.3V

However, Current sense circuit not fully understood.
Practically, measuring current output of battery without
a 5m Sense resistor and differential amplifier is challenging.

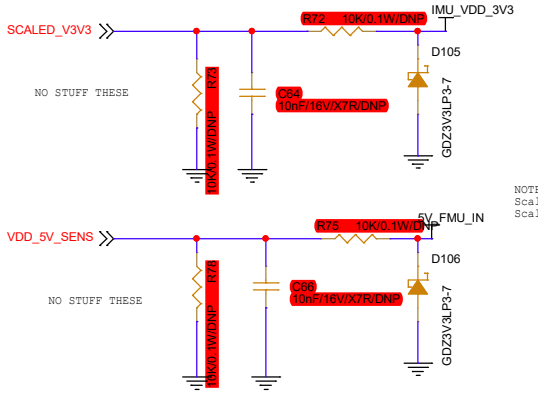
Investigation necessary

Battery 1



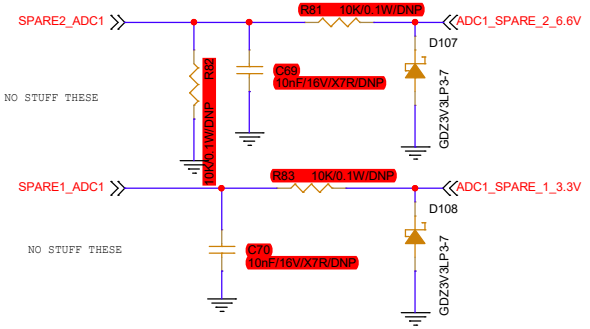
Note: Sensing of Voltage and Current from Power Modules to be confirmed by Client

Core voltage detection



NOTE:
Scaled 3.3V outputs aprox 1.7V Max
Scaled 5V outputs 2.5V Max

ADC 3.3V & 6.6V



DIGITAL SHARK TECHNOLOGY		
Title: ADC & Sensing		
Size: B	Document Number: 1.0	Rev: 1.1
Date: Friday, November 14, 2025	Sheet 9 of 9	